PROJECT CASE STUDY

Bank Customer Churn Prediction

Classical ML + Neural Network Comparison

Category	Status	Year
Applied ML	Completed	2025

A structured customer-risk modeling project comparing boosting pipelines with MLP and feed-forward neural-network approaches.

2	Pipeline	Business
model families	reproducible prep	retention framing

Context and problem

Banks need interpretable and reproducible ways to identify customers who are likely to leave, so retention work can be targeted earlier.

Strategy and implementation

- Cleaned and encoded customer attributes through reusable preprocessing pipelines.
- Compared boosting algorithms with MLP and FNN models.
- Evaluated models using classification metrics rather than accuracy alone.
- Framed predictions in terms of retention decisions and customer-risk patterns.

My role

End-to-end data preparation, model development, comparison, evaluation, and interpretation.

Technology stack

Python | Pandas | scikit-learn | Boosting | MLP | FNN | Matplotlib

Results and evidence

- Built a repeatable model-comparison workflow.
- Demonstrated both classical and neural approaches on the same business problem.
- Created a portfolio-ready example of structured tabular ML.

Key takeaways

- The best churn model must be judged with business-relevant errors.
- Strong preprocessing often matters more than increasing neural-network depth.

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